Data-Driven Optimization and Scalability of The Diabetes Care Medical and General Stores in Retail Pharmacy sector

A mid-term submission report for the BDM capstone Project

Submitted by

Name: **Arya Mulay**

Roll number: **23F3002123**

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents Page No.

1. Executive Summary and Title

Project Title: Data-Driven Optimization and Scalability of The Diabetes Care Medical and General Stores in Retail Pharmacy sector.

The Pharmaceutical firm, Diabetes Care Medical and General Stores, established in 2010, is run by the owner and one assistant. They have overcome financial hurdles and built strong supplier relationships over the past 15 years. With steady profits, they have strong potential to scale up but struggle to do it by not utilizing the credit period given by the suppliers.

Two significant operational issues that Diabetes Care Medical and General Stores face are examined in this mid-term report: the inability to open new retail locations despite strong profits and the regular inventory imbalances brought on by antiquated systems and poor stock visibility.

Ten months of primary data (April 2024–January 2025) were gathered from the company's internal purchase, sales, GST, and inventory records in order to examine these problems. To enable precise and effective analysis, the data was cleaned, formatted, and arranged across two spreadsheets with well-defined metadata for every sheet and column.

Utilization of supplier credit was the main focus of the analysis's first section. The pharmacy had roughly ₹11.45 lakhs of unused capital locked up in credit cycles, according to estimates of trapped capital and the Weighted Average Credit Period (WACP). To investigate funding options for a new branch, a working capital simulation was carried out. Without the need for outside loans, using this trapped capital in conjunction with retained profits provided the highest return on investment (46.88%).

Inventory control was the subject of the second section. Tighter control is required in areas where the top 50% of medications contribute nearly 79% of the total inventory value, according to ABC analysis. Overstocked and quickly-moving items were identified by Inventory Turnover Ratio (ITR) analysis, which improved reorder tactics and decreased waste.

The findings demonstrate that the pharmacy can scale operations more effectively while lowering operational and financial risks by improving credit utilization and practicing smarter inventory management.

2. Proof of Originality

Primary Data: The primary data of 10 months (april 2024 to january 2025) is collected from the pharmacy firm arranged in two spreadsheets as given below:

- Purchase and GST details dataset (records from 01-04-24 to 31-01-25): {Click Here}
- Sales and Inventory details dataset (records from 01-04-24 to 31-01-25): {Click Here}

Pharmacy Firm Images: Google Drive Link: {Click Here}

Interaction Video: I have uploaded the interaction video on youtube along with subtitles in English. **Link:** {Click Here}

Letterhead from the organization (Diabetes Care Medical and General Stores):
Link: {Click Here}

3. Metadata and Descriptive Statistics

3.1. Metadata:

The metadata for Purchase and GST dataset, the google sheet dataset file{Click Here} along with its respective sheet name is displayed in the below Table 1:

TABLE 1: METADATA FOR PURCHASE AND GST DATASET

Sheet name	Columns	Data type	Units	Description
DATA SUMMARY (It contains all purchase and gst records of 10 months from April 2024 to January 2025)	Date	date	monthwise date	Represents the month and year of Purchase data
	Purchase	float	Rupees	The total purchase amount is over 10 months.
	GST Paid	float	Rupees	The total GST paid by suppliers over 10 months.
	Total Purchases	float	Rupees	The total purchase including the gst amount paid over 10 months.
	Sale	float	Rupees	The total sale amount is over 10 months.
	GST Collected	float	Rupees	The total GST collected from consumers over 10 months.
	Total Sales	float	Rupees	The total sale including the gst amount collected over 10 months.
	Gross Profit	float	%	Raw Profit computed by only including sale amount and purchase amount
	Net Profit	float	%	Net Profit computed by using total sale and total profit along with other variables like stationary, employee salary and database maintenance
1. 'APRIL2024' 2.'MAY2024' 3.'JUNE2024' 4.'JULY2024' 5.'AUGUST2024' 6.'SEPTEMBER2024' 7.'OCTOBER2024' 8.'NOVEMBER2024' 9. DECEMBER2024' 10. 'JANUARY2025'	Date	date	dd/mm/yyyy	Represents the month and year of each transaction
	Invoice No.	string	-	Unique invoice number issued by supplier
	Name of Party	string	-	Name of the vendor/supplier
	GSTR No.	string	-	Registered GST number of the vendor/supplier
	Purchase Amount	float	Rupees	Purchase value excluding GST
	GST Value 5%	float	Rupees	GST amount calculated at 5% slab
	GST Value 12%	float	Rupees	GST amount calculated at 12% slab
	GST Value 18%	float	Rupees	GST amount calculated at 18% slab

The metadata for Sales and Inventory dataset, the google sheet dataset file{Click Here} along with its respective sheets is displayed in the below Table 2:

TABLE 2: METADATA FOR SALES AND INVENTORY DATASET

Sheet name	Columns	Data type	Units	Description
1.'APRIL2024' 2.'MAY2024'	Medicine name	string	-	Name of the medicine

3.'JUNE2024' 4.'JULY2024' 5.'AUGUST2024' 6.'SEPTEMBER2024' 7. 'OCTOBER2024' 8.'NOVEMBER2024' 9. 'DECEMBER2024' 10. 'JANUARY2025'	Price per Strip	float	Rupees	Selling price per strip (1 strip = 10 tablets/capsules)
	Opening Stock	integer	strips	Number of strips in stock at the beginning of the month
	Purchase Quantity	integer	strips	Number of strips purchased during the month
	Sale Quantity	integer	strips	Number of strips sold during the month
	Closing Stock	integer	strips	Number of strips in stock at the end of the month

3.2. Descriptive Statistics:

Below Table 3 and Table 4 both provide descriptive statistics for the whole period of 10 months, also I have provided individual month statistics in the dataset sheet.

TABLE 3: Descriptive Statistics for overall 10 month purchase data

Descriptive Statistics for "PURCHASE DATA SUMMARY"					
Features/ Parameters	**GST paid (in ₹)**	**Total Purchase (in ₹)**	**GST collected (in ₹)**	**Total Sales (in ₹)**	**Net Profit (in %)**
Count	10	10	10	10	10
Mean	₹49,527.42	₹559,180.22	₹67,948.15	₹696,714.02	19.37%
Standard Deviation	₹6,475.32	₹74,741.59	₹6,653.83	₹72,565.43	5.00%
Variance	₹41,929,743.40	₹5,586,304,870.73	₹44,273,417.12	₹5,265,741,329.82	0.25%
Minimum Value	₹40,503.04	₹462,685.04	₹53,770.51	₹563,670.00	10.82%
Maximum Value	₹62,987.80	₹717,334.80	₹77,586.09	₹822,375.00	27.92%
25th Percentile	₹45,300.09	₹513,954.59	₹64,524.68	₹654,290.77	16.43%
Median	₹48,757.77	₹545,747.77	₹69,049.71	₹699,273.50	19.43%
75th Percentile	₹52,775.90	₹591,429.65	₹71,654.46	₹744,201.87	22.03%
Mode	-	-	-	-	-

Table 3 Inference: The average monthly total purchase was ₹5.59 lakhs and total sales averaged ₹6.97 lakhs, yielding net profit which includes the employee salary, stationary cost as well as software maintenance averaging it to 19.37%.

Features/ Parameters	**Date/Period**	**Opening Stock**	**Purchase Qty**	**Sale Qty**	**Closing Stock**
Count	Apr - 24 to Jan - 25	20	20	20	20
Mean	Apr - 24	89.40	38.75	61.55	66.60
	May - 24	66.60	54.50	57.70	63.45
	Jun - 24	63.45	50.95	58.55	55.85
	Jul - 24	55.85	57.50	57.85	55.50
	Aug - 24	55.50	61.90	62.90	54.50
	Sep - 24	54.50	60.25	66.25	48.50
	Oct - 24	48.50	65.25	65.25	48.50
	Nov - 24	48.50	63.50	63.30	48.70
	Dec - 24	48.70	63.50	63.55	48.65
	Jan - 25	48.65	66.00	60.85	53.80
Standard Deviation	Apr - 24	171.89	74.51	131.97	114.30
	May - 24	114.30	129.40	142.66	100.85
	Jun - 24	100.85	118.33	134.37	85.31
	Jul - 24	85.31	140.39	133.54	91.41
	Aug - 24	91.41	150.91	140.56	100.24
	Sep - 24	100.24	139.53	145.90	93.10
	Oct - 24	93.10	150.17	144.10	98.94
	Nov - 24	98.94	138.93	142.09	95.69
	Dec - 24	95.69	138.98	141.97	93.00
	Jan - 25	93.00	149.94	132.74	109.96

Features/ Parameters	Date/Period	Opening Stock	Purchase Qty	Sale Qty	Closing Stock
Median	Apr - 24	44.00	20.00	31.50	36.00
	May - 24	36.00	25.00	28.00	38.50
	Jun - 24	38.50	22.50	28.00	32.00
	Jul - 24	32.00	25.00	28.00	32.50
	Aug - 24	32.50	25.00	32.00	30.00
	Sep - 24	30.00	35.00	34.00	30.50
	Oct - 24	30.50	37.50	34.00	29.00
	Nov - 24	29.00	32.50	36.00	28.00
	Dec - 24	28.00	35.00	35.00	26.50
	Jan - 25	26.50	35.00	32.00	30.50
Mode	Apr - 24	16	5	47	88
	May - 24	88	5	28	9
	Jun - 24	9	20	24	26
	Jul - 24	26	25	16	8
	Aug - 24	8	25	38	30
	Sep - 24	30	40	34	37
	Oct - 24	37	40	42	31
	Nov - 24	31	30	36	28
	Dec - 24	28	35	36	25
	Jan - 25	25	25	32	13

Table 4 Inference: Average sale quantity per month ranged from 57.70 to 66.25 strips, with Sep 2024 as the highest sales quantity average of 66.25 strips in the month, while Nov 2024 with highest median of 36.00 strips per month.

4. Detailed Explanation of Analysis Process

The data analysis is divided into the following subparts

4.1. Data Collection: This primary data has been collected in two parts from April 2024 to January 2025 (10 months), one is the raw purchase and sales data obtained by the excel files as provided by the pharmaceutical company and the other on the sales and inventory data consists of major 20 purchased medicines obtained via the Green Cross Database Software as provided by the pharmaceutical company and discussion with the owner. After collecting the above mentioned data I have systematically organized both the datasets which will further aid in comfortable viewing and analysis of the data.

4.2. Data Cleaning and Preprocessing: The primary data divided into two separate google sheets having separate sheets of each lacked the required formatting and improper data presentation like the GST value units in rupees symbol(₹), the profit and net profit in percentage symbol(%), the quantities of medicines in the required strips(1 strip = 10 tablets), some required the quantities to be in tubes/ vials and the GSTR no. which is alphanumeric text. The unwanted and incomplete columns and rows have been removed from the dataset. All the required formatting is applied to the columns in all sheets of both the files as mentioned in the metadata section (Please refer Table 1 and Table 2 for metadata).

This particular cleaning process made it easier to further work and perform analysis on the datasets according to my problem statements.

4.3. Analysis Process:

For the **problem statement 1**, about scaling up of the organization the analysis are as follows:

4.3.1. **Supplier Credit Utilization Analysis:** The main objective of this analysis is to quantify the unused credit period from the distributors and identify opportunities for raising capital to scale up the business.

Firstly lets calculate Weighted Average Credit Period (WACP) using the mathematical equation: *WACP=∑(Purchase Amount × Credit Days Used)/∑Purchase Amount*

Next to get the trapped capital for 10 months using the mathematical equation:

Let C_i = Credit Days Used for purchase amount i and Let A_i = Purchase Amount from i th supplier. So, *Trapped Capital of 10 months=∑(Ai×Unused Daysi)/306*

4.3.2. **Working Capital Simulation:** The main motive of this analysis is to calculate how utilizing the unused supplier credit period can fund one new branch using 10-month data and project ROI in quantitative terms.

So, total Trapped Capital for 10 months amounting to ₹1,145,626.98 could have been used to fund a new branch of the pharmaceutical firm. Suppose the total Setup Cost for opening a new branch would be about ₹1,600,000.00 including the rent area, employee salary, hardware components, database softwares, stationary, minimum required drug stocks and other operational costs.

Now to project Return on Investment(ROI) annually for the new branch, let's assume the annual profit of the new branch would be about 60% that of the annual profit of the main branch, which would be about ₹7,50,000.

Mathematical Equation: $ROI = (Annual\ Net\ Profit\ /\ Total\ Investment)X100$

For the **problem statement 2** about inventory data, analysis are as follows:

4.3.3. **ABC Analysis:** ABC Analysis helps categorize inventory items based on their annual consumption value, but here we have 10 month data so we categorize inventory based on 10 month consumption value. This helps the firm to:Identify high-priority medicines that contribute most to total inventory cost.

By this analysis you can identify the high consumption value items (Class A), so that the firm can set tighter controls, reduce stockouts, and prioritize supplier management, while the class C items can have a bit of relaxed monitoring and maintain a low stock buffer.

To start with the analysis we need the total consumption value (in ₹) for each medicine over complete 10 months by using the mathematical equation:

*Total Consumption Value= ∑(Price per strip*Total Sale Quantity)*

Then for each item calculation of percentage of consumption value is given by the mathematical equation:

$\%\ of\ Total = (Item\ Consumption\ Value/\ Grand\ Total\ Consumption\ Value\ )\times 100$

Next to compute the cumulative %, the data must be arranged in descending order of the consumption value. Next we classify the items into A,B and C classes respectively using the Pareto Principle (80/20 rule). The Top 80% value comes under class A, followed by next 15% under class B and the last 5% covered under class C.

4.3.4. **Inventory Turnover Ratio (ITR) Analysis:** The goal of the Inventory Turnover Ratio (ITR) analysis is to evaluate the current efficiency of inventory in terms of stock management and utilization for each of the 20 medicines available in the dataset.

To evaluate this, using the mathematical equation for Inventory Turnover Ratio (ITR) for each of the 20 medicines: *ITR=Total Sale Quantity (in strips)/Average Inventory Stock (in strips)*, where the Average Inventory Stock is computed by: *Average Inventory=(Opening Stock+Closing Stock)/2.*

This analysis is better than looking at just sales and quantity in isolation, because the sales and inventory behaviour of each medicine captures its realistic stock efficiency.

Please refer the google excel sheet for all the above mentioned Analysis identifiable by their respective sheet name - {Click here}

5. Results and Findings

For the analysis of problem statement 1:

5.1. Supplier Credit Utilization Analysis:

By using the WACP(used) formula and referring to the table we get WACP(used) as 112.62 days whereas the WACP(unused) is 64.08 days which shows that the firm has a huge potential to increase the capital by utilizing more of its credit period in purchase payment to the suppliers.

The total Trapped Capital for the 10 months data provided by the firm amounts to ₹1,145,626.98 by using the formula of Trapped Capital as stated in the detailed analysis section.

The Supplier Credit Utilization Analysis resulted in the below two charts:

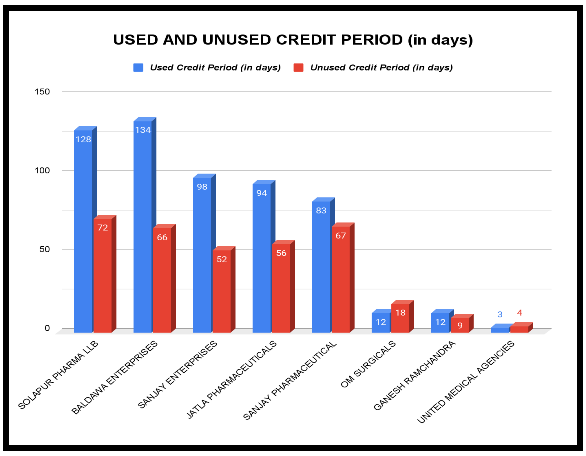


Fig 2: Stacked Bar Chart

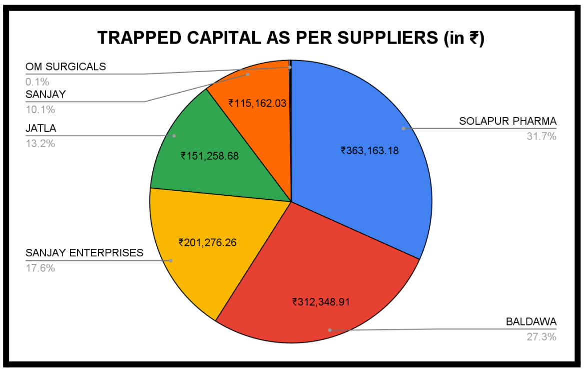


Fig 3: Standard Pie Chart

Insights:

a. The Stacked Bar Chart (Fig 2) clearly shows the difference between used and unused credit periods for each supplier. It highlights that the pharmacy has strong potential to unlock capital by better utilizing the available credit days. Strengthening supplier relationships can help convert this unused credit into investment for scaling up.

b. The Pie Chart (Fig 3) shows how much trapped capital is tied to each supplier due to underutilized credit periods. The highest amounts are with Solapur Pharma LLB at ₹363,163.18 and Baldawa Enterprises at ₹312,348.91. Others like Sanjay Enterprises, Jatla Pharmaceuticals and Sanjay Pharmaceuticals also hold sizable amounts worth targeting.

5.2. Working Capital Simulation:

The <u>total trapped capital</u> if utilized covers up to <u>71.6%</u> of the total branch setup cost.Therefore <u>additional funding needed</u> = ₹1600000 - ₹1145626.98 = <u>₹454373.02</u>. This amount can either be taken from the profit generated by the main branch or it can be fully loaned out. So the <u>ROI</u> calculated by the mathematical equation for different scenarios is as follows:

<u>Scenario 1</u>: Using trapped capital and profits from main branch, ROI = 46.88%

<u>Scenario 2</u>: Using trapped capital and loan for the remaining amount, ROI = 44.60%

<u>Scenario 3</u>: Using loan for the complete capital funding, ROI = 38.88%

The Working Capital Simulation resulted in following graphs:

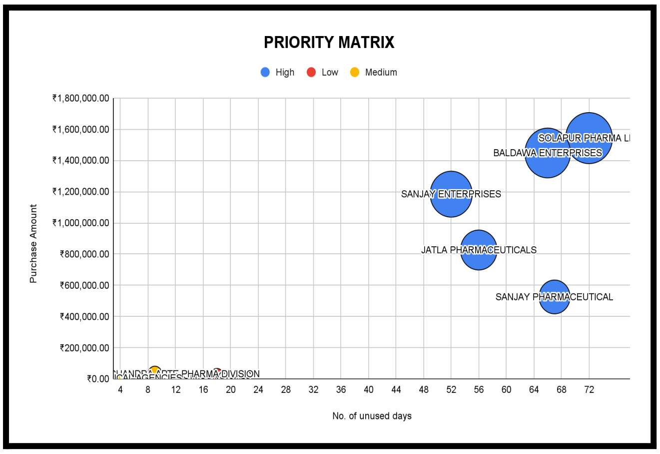


Fig 4: Bubble Chart

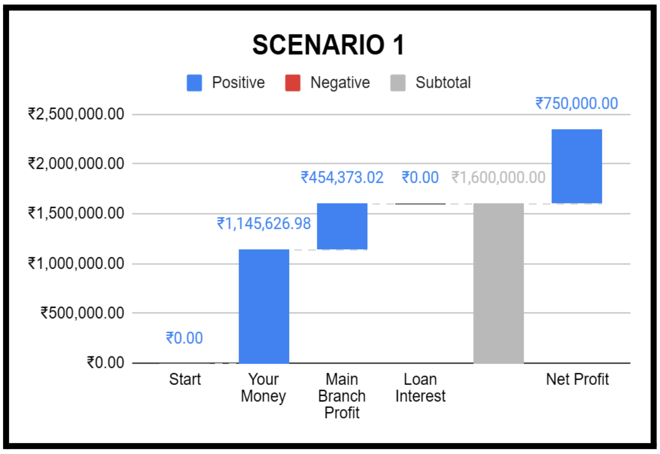
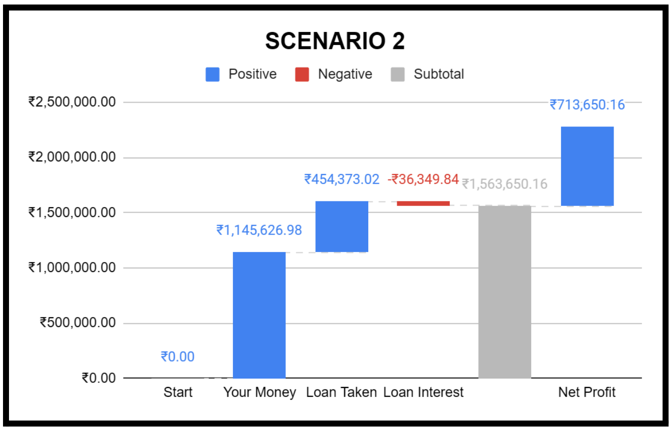


Fig 5: Waterfall Chart (Scenario 1)



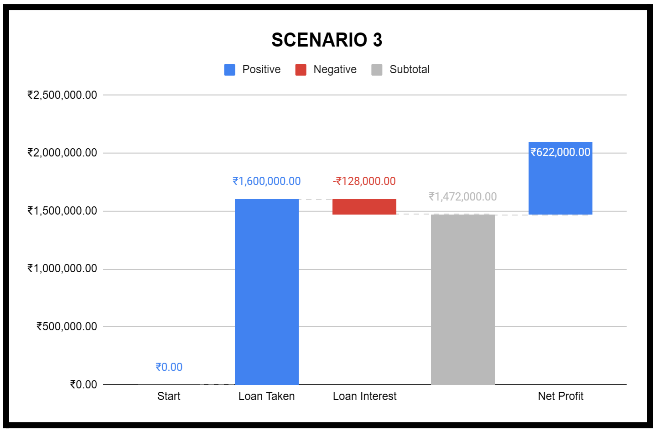


Fig 6: Waterfall Chart (Scenario 2)

Fig 7: Waterfall Chart (Scenario 3)

Insights:

a. The Bubble chart (Fig 4) indicates the priority matrix which indicates the maximum amount of trapped capital based on the number of unused credit days as X - axis, the Y - axis representing the purchase amount and Trapped Capital as the size of the bubble. We observe that the Solapur Pharma LLB and Baldawa Enterprises are one of the highest priority during the negotiation of credit period as they have high unused days and massive trapped capital.

b. Now coming to the waterfall chart for Scenario 1 (Fig 5) which ensures the highest ROI (46.88%) by using the complete trapped capital which covers 71.6% of the total capital investment of ₹1,600,000. The remaining amount can be funded by cutting down profits gained from the main branch, therefore no loan interest and ideal case to boost the ROI.

c. The next Scenario 2 (Fig 6) where we use the trapped capital along with loaning out the remaining amount rather than cutting down the profit gained from main branch, then the remaining amount (₹454,400) can be completely loaned resulting in a very minimalistic ROI dropdown to 44.60%. This calculation is done considering the loan interest is about 8%p.a also lowering the net profit from ₹750,000 to ~ ₹713,650.

d. The next waterfall chart about Scenario 3 (Fig 7) which is least suited is without using the trapped capital, the complete funding of ₹1,600,000 resulting in ROI of 38.88%, along with chopping down the assumed profit of the new branch from ₹750,000 to ₹622,000.

Conclusion: The pharmacy firm should make full use of the supplier credit period, especially from suppliers like Solapur Pharma LLB and Baldawa Enterprises who top the priority list. For funding the new branch, Scenario 1 is the best choice as it uses trapped capital and profits from the main branch, avoiding the need for loans.

For the analysis of problem statement 2:

5.3. ABC Analysis:

The main reason to choose ABC Analysis is it is one of the most efficient inventory prioritization techniques used in supply chain and pharmacy inventory systems. Unlike other equal-priority systems it focuses on value criticality and gives disproportionate importance to high-impact items rather than perishability.

The ABC Analysis resulted in the following Preto Chart:

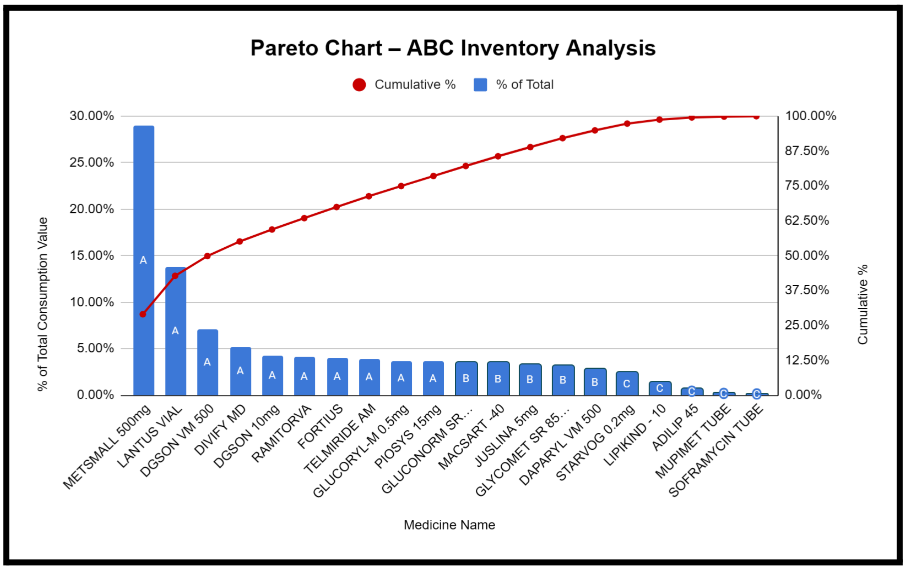


Fig 8: Pareto Chart - ABC Inventory Analysis

Insights:

a. The Pareto Chart provides the data labels for their respective class (A,B,C) giving us a broad view of the inventory management.

b. The Class A items, contributing the highest share (~78.57%) of the overall consumption value, have all the high moving stock consisting of 50% of the medicines from the data available. These sets of items need proper management, tight replenishment and frequent observation as they are critical for minimizing capital lock in and avoiding shortage of supplies.

c. The Class B items, contributing a moderate share (~16.35%) of the overall consumption value, consist of 25% of the total medicines. These items require periodic monitoring and balanced stock control. They are not as critical as Class A, but must not be neglected as they can transition to higher priority based on changing demand or prescription patterns.

d. The Class C medicines contribute the least (~5%) to the total consumption value and also form 25% of the total items. These are low-priority items that can be ordered in larger, infrequent batches, and may not need close monitoring. Their limited contribution suggests opportunities to optimize inventory space by potentially phasing out underperforming items.

5.4. Inventory Turnover Ratio (ITR) Analysis :

The choice of this analysis is justifiable as it helps to classify inventory items into overstocked or slow moving, efficiently maintained and fast moving or at the risk of stockouts based on how efficiently they move.

The Inventory Turnover Ratio (ITR) Analysis resulted in the following bar chart:

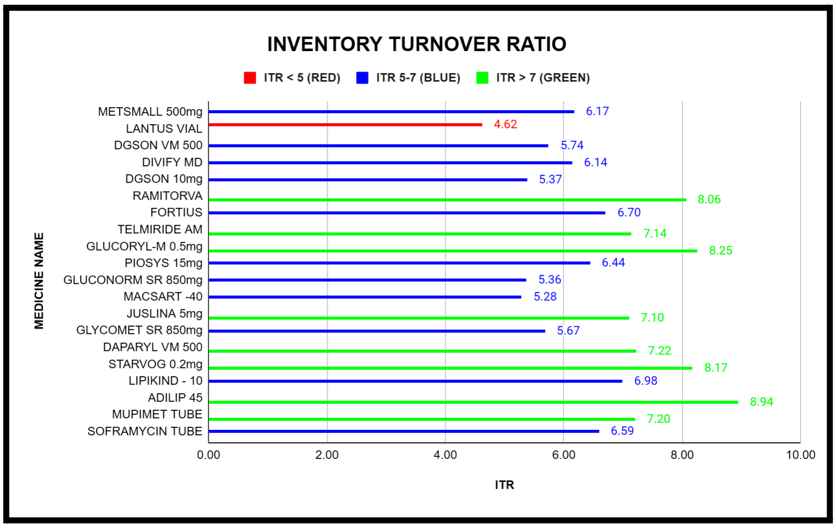


Fig 9: Column Chart - ITR Analysis

Insights:

a. The column chart is color-coded by its ITR values into Low Turnover depicted by red color, then comes Balanced Turnover depicted by blue color series and then High Turnover depicted by green color.
b. The Low Turnover Medicines whose ITR is less than 5 represent that the medicine is overstocked and indicates excessive capital tied up and risk of expiry leading to wastage.
c. The medicines falling within the ideal ITR range of 5 to 7 indicate a balanced and efficient inventory management system. These items are neither overstocked nor at risk of frequent stockouts. Their stock levels align well with demand, suggesting that the current reordering frequency and quantity are appropriate and should be maintained consistently.
d. Medicines with an ITR greater than **7**, are fast-moving items that may be at risk of stockouts if not restocked in time. These drugs have strong demand but may lack sufficient buffer stock. Therefore, it is advisable to increase the reorder frequency for such items to avoid disruptions in patient care.

Conclusion: The combined ABC and ITR analysis shows that focusing on high-value fast-moving medicines can greatly improve inventory efficiency. By adjusting stock levels for both fast and slow movers, the pharmacy can reduce capital blockage, avoid stock issues, and maintain better overall control.